

Algebra 4

Vincent Wang

October 2025

1 Free Abelian Groups

Any subgroup N of a free abelian group L of rank n has rank $\leq n$.

Proof: Treat L as $\mathbb{Z}^n$. We induct on n . When $n = 0$, the result holds. Suppose every element of N has zero first coordinate. Then we can embed $N \hookrightarrow \mathbb{Z}^{n-1}$, so N is free of rank at most $n-1 \leq n$. Else, let $g = \min\{\pi_1(x) : \pi_1(x) > 0, x \in N\}$ (by Well-ordering of $\mathbb{Z}^+$). Pick $f \in N$ with $\pi_1(f) = g$.

Denote by $N' = \{x \in N : \pi_1(x) = 0\}$. $N' < N$ since for $x, y \in N'$, $\pi_1(x+y) = \pi_1(x) + \pi_1(y) = 0$ and $\pi_1(-x) = -\pi_1(x) = 0$. Since g divides any first coordinate, $N < \langle f \rangle + N'$. Since $\langle f \rangle, N' < N$, we have $\langle f \rangle + N' < N$, so $\langle f \rangle + N' = N$. N' can be embedded in $\mathbb{Z}^{n-1}$, so N' is free of rank $\leq n-1$ by the induction hypothesis. If $\{f_1, f_2, \dots, f_r\}$ is a basis of N' , then $\{f, f_1, f_2, \dots, f_r\}$ is a basis of N , so N is of rank at most n .

Let C be an integer $n \times m$ matrix. Then C can be transformed by integer elementary operations into a diagonal matrix with elements $u_1, u_2, \dots, u_k$, where $k = \min(n, m)$ and $u_i \mid u_{i+1}$ for $i \in [k-1]$.

(0). If $C = 0$, we're done. Else, $c_{ij} \neq 0$ for some i, j . Row and column swap so that $c_{11} \neq 0$ and multiply by -1 if needed so $c_{11} > 0$.

(1). For each element c_{i1} (for $i > 1$) in the first column, write by Division algorithm $c_{i1} = c_{11}q_i + r_i$ for $0 \leq r_i < c_{11}$. If $r_i \neq 0$ for some i , then reduce by operation (ii) so that the i th row starts with r_i and then swap rows $1, i$. If $r_i = 0$ for all i , move to step (2).

(2). Similarly, for each element c_{1j} of the first row, write $c_{1j} = c_{11}p_j + s_j$. If $s_j \neq 0$ for some j , eliminate and swap so that column 1 starts with s_j and go to step (1). If $s_j = 0$ for all j , move to step (3).

(3). All entries in row and column 1 are zero except for $c_{11} > 0$. Suppose for some $i, j > 1$ that $c_{11} \nmid c_{ij}$. Add row i to row 1. Since the first entry of row i is 0, in the new matrix, c_{11} has the same value as before and c_{1j} equals

the old value c_{ij} (which also equals the new c_{ij}). Then go to step (2). Else, terminate at (4).

(4). At this state, all entries in row and column 1 are zero except for $c_{11} > 0$ and for all $i, j > 1$, we have $c_{11} \mid c_{ij}$.

The algorithm above terminates because c_{11} is ensured to be positive and decreasing, and a descending chain of positive integers must be finite. By induction, C has the desired form as a diagonal matrix.

Let $\{e_i\}_{i=1}^n$ be a basis of L , $\{f_j\}_{j=1}^m$ a basis of N for $m \leq n$. Since $\{e_i\}$ generates L , for each f_j let $f_j = \sum_{i \in [n]} c_{ij} e_i$, so C is a $n \times m$ matrix with each column representing an element f_j . The previous work shows that there exist invertible integer matrices L, R such that $L^{-1}CR = D$, where D is diagonal with the shape described above. That is, for the change of bases $(e'_i) = (e_i)R$ and $(f'_j) = (f_j)L$, the matrix of the basis elements (f'_j) of N in the basis (e'_i) of L is D .

Let A be a finitely generated Abelian group, so $A = \langle a_1, a_2, \dots, a_n \rangle$ for some a_i . Let F be the (up to isomorphism) abelian group of rank n on the generators $f_1, f_2, \dots, f_n$. There exists a homomorphism $\phi : F \rightarrow A$ such that $f_i \mapsto a_i$ by the extension property of F . Since A is generated by the $\{a_i\}$, ϕ is surjective.

By the First Isomorphism theorem, $A \cong F/N$, where $N = \ker \phi$. Since F is abelian, N is no more than a subgroup, so A can be viewed as the quotient of a free abelian group.

Pick bases $\{e_i\}$ and $\{f_j\}$ of F, N respectively such that the corresponding matrix of $\{f_j\}$'s in the $\{e_i\}$'s has shape D , so $f_i = u_i e_i$ for $i \in [m]$ (the columns of D are nonzero since F has a basis of size m). Then $F/N \cong \mathbb{Z}_{u_1} \oplus \mathbb{Z}_{u_2} \oplus \dots \oplus \mathbb{Z}_{u_m} \oplus \mathbb{Z}^{n-m}$.

Hence, rephrasing notation and refactoring, for any finitely generated Abelian group G , we may write $G = (\bigoplus G(p)) \oplus \mathbb{Z}^r$ for p -groups $G(p)$ and $r \geq 0$.

Suppose $G \cong H$. Then $G/T(G) \cong H/T(H)$. Let $\phi : G \rightarrow H$ be an isomorphism, $p_G : G \rightarrow G/T(G)$, $p_H : H \rightarrow H/T(H)$ the projection maps. Consider the composition $G \xrightarrow{\phi} H \xrightarrow{p_H} H/T(H)$. If $x \in T(G)$, then $\phi(x) \in T(H)$, so this map descends to a map $\varphi_G : G/T(G) \rightarrow H/T(H)$. Similarly, we have a map $\varphi_H : H/T(H) \rightarrow G/T(G)$ (instead using the inverse map $\phi^{-1} : H \rightarrow G$).

Suppose $x \in G$. Then $\varphi_H \varphi_G(x + T(G)) = \varphi_H(\phi(x) + T(H)) = \phi^{-1} \phi(x) + T(G) = x + T(G)$, so $\varphi_H \varphi_G = \text{id}_{G/T(G)}$. Similarly, $\varphi_G \varphi_H = \text{id}_{H/T(H)}$, so φ_G, φ_H are inverses.

Clearly, if $G \cong (\bigoplus G(P)) \oplus \mathbb{Z}^r \cong (\bigoplus G'(p)) \oplus \mathbb{Z}^{r'}$, then $\mathbb{Z}^r \cong G/T(G) \cong$

$G'/T(G') \cong \mathbb{Z}^{r'}$. Since $\mathbb{Z}^r, \mathbb{Z}^{r'}$ are free of rank r, r' and rank is unique, $r = r'$.

If $G \cong H$, then clearly $T(G) \cong T(H)$, so $\bigoplus G(p) \cong \bigoplus H(p)$.

Suppose $G \cong H$ via the isomorphism ϕ . Denote by $G(p) = \{x \in G : p^k x = 0 \text{ for some } k \geq 0\}$. Denote by $H(p)$ similarly. Consider first the restriction of ϕ to $G(p) \rightarrow H$. Suppose $x \in G(p)$, so $p^k x = 0$ for some $k \geq 0$. Then $\phi(p^k x) = p^k \phi(x) = 0 \rightarrow \phi(x) \in H(p)$. Hence, we may restrict ϕ to $\phi' : G(p) \rightarrow H(p)$.

Suppose $y \in H(p)$. Since ϕ is surjective, there exists $x \in G$ such that $\phi(x) = y$. Since $y \in H(p)$, there exists $k \geq 0$ such that $p^k y = 0$. Then $\phi(p^k x) = p^k \phi(x) = p^k y = 0 \rightarrow p^k x = 0$ (since ϕ is injective), so $x \in G(p)$. It follows that ϕ' is also surjective and therefore an isomorphism (injectivity is given).

Then it suffices to consider $G \cong H$ for finite p -groups G, H . Then consider $G/G^p \cong H/H^p$ to force equality of order types.

2 Euclidean Domains and Principal Ideal Domains

Let R be a PID and $u \in R - R^\times - 0$. Then the following are equivalent:

- (i). $R/\langle p \rangle$ is a field.
- (ii). $R/\langle p \rangle$ is an integral domain.
- (iii). p is prime.
- (iv). p is indecomposable.

Suppose (i). Then (ii) holds since all fields are integral domains. Then $\langle p \rangle$ is a prime ideal, ie if $ab \in \langle p \rangle$, then either $a \in \langle p \rangle$ or $b \in \langle p \rangle$, so p is prime. Now suppose $p = xy$ for some $x, y \in R - R^\times - 0$. Since $p \in \langle p \rangle$, we have $x \in \langle p \rangle$ or $y \in \langle p \rangle$. WLOG $x \in \langle p \rangle$, so $x = px_1$ for some $x_1 \in R$. Then $px_1 y = p$. Since $p \neq 0$ and R has no zero divisors, $x_1 y = 1$, so $x_1, y \in R^\times$, absurd.

Finally, assume that p is indecomposable. I claim that $\langle p \rangle$ is a maximal ideal. Suppose otherwise. Since R is a PID, we may assume that $\langle p \rangle \triangleleft \langle q \rangle \triangleleft R$, where inclusions are proper. Since $p \in \langle q \rangle$, there exists $x \in R$ such that $p = qx$. If $x = 0$, then $p = 0$, absurd. If $x \in R^\times$, then $\langle p \rangle = \langle q \rangle$, absurd. Hence, $x \in R - R^\times - 0$. If $q = 0$, then $\langle p \rangle = 0$, absurd. If $q \in R^\times$, then $\langle q \rangle = R$, absurd. Hence, $q \in R - R^\times - 0$, so p is decomposable, absurd.

Since $\langle p \rangle$ is maximal, $R/\langle p \rangle$ is a field, so (iv) implies (i). The PID properties allow us to assume $\langle q \rangle$ has this shape.

If p is prime and $a \notin \langle p \rangle$, then $[a] \neq [0]$. Since $R/\langle p \rangle$ is a field, pick $x \in R$ such that $[a][x] = [1]$, or that $ax - 1 \in \langle p \rangle$, ie for some $y \in R$ we have $ax + py = 1$.

Let E be a Euclidean domain. Then E is a PID. Suppose $I \triangleleft E$. If $I = 0$, we're done. Else, $I \neq 0$ so the set $\{N(x) : x \in I - 0\} \subset \mathbb{Z}^+$ is nonempty. By well-ordering of $\mathbb{Z}^+$, pick $d \in I$ with minimal norm. Suppose $x \in I$. Write $x = dq + r$ with $N(r) < N(d)$. If $r \neq 0$, then $N(r) \neq 0$ and $r = x - dq \in I$ has smaller norm than d , contradicting minimality of $N(d)$. Hence, $r = 0$ and $x = dq \in \langle d \rangle$, ie $I \subset \langle d \rangle$. Conversely, since $d \in I$, we have $\langle d \rangle \subset I$, so $I = \langle d \rangle$ is principal.

$\mathbb{Z}[i]$ is a Euclidean domain. Suppose $a + bi, c + di \in \mathbb{Z}[i]$ with $c + di \neq 0$. Then $\frac{a+bi}{c+di} = \alpha + \beta i \in \mathbb{Q}[i]$. Pick $x + yi$ such that $|\alpha - x|, |\beta - y| \leq \frac{1}{2}$, so

$$N(a + bi - (c + di)(x + yi)) = N(c + di)N\left(\frac{a + bi}{c + di} - (x + yi)\right) \leq \frac{1}{2}N(c + di) < N(c + di).$$

Take $r + si = a + bi - (c + di)(x + yi) \in \mathbb{Z}[i]$.

Hence, $\mathbb{Z}[i]$ is a Euclidean domain and therefore a PID.

Suppose E is a Euclidean domain that is not a field. Then E has universal side divisors. Let $u \in R - R^\times - 0$ be N -minimal. Suppose there exists $x \in E$ such that $x = qu + r$ with $r \in R - R^\times - 0$. Then $N(r) < N(u)$, contradicting minimality of $N(u)$. Hence, $r \in R^\times \cup 0$.

$\mathbb{Z}[x]$ is (an integral domain that is) not a PID. I claim that $\langle 2, x \rangle$ is not principal. Suppose otherwise, so $\langle 2, x \rangle = \langle f \rangle$ for some $f \in \mathbb{Z}[x]$. Then $f \mid 2, x$, so $f \in \{\pm 1, \pm 2\}$, $f \in \{\pm 1, \pm x\}$, so $f = \pm 1$. Either way, $\langle f \rangle = \mathbb{Z}[x]$, absurd.

$R = \mathbb{Z}[\sqrt{-5}]$ is an integral domain that is not a PID. Suppose FTSOC $I = \langle 3, 2 + \sqrt{-5} \rangle = \langle a + b\sqrt{-5} \rangle$. Since the norm $N : \mathbb{Z}[\sqrt{-5}] \rightarrow \mathbb{Z}_{\geq 0}$ given by $N(a + b\sqrt{-5}) = a^2 + 5b^2$ is multiplicative, we have $a^2 + 5b^2 \mid 9$, since $N(3) = N(2 + \sqrt{-5}) = 9$. When $N(a + b\sqrt{-5}) = 1$, $a = \pm 1$. When $N(a + b\sqrt{-5}) = 3$, there are no solutions. When $N(a + b\sqrt{-5}) = 9$, the solutions are $\pm 3, \pm 2 \pm \sqrt{-5}$.

Let $x + y\sqrt{-5}, u + v\sqrt{-5} \in R$ such that $(x + y\sqrt{-5})(a + b\sqrt{-5}) = 3$ and $(u + v\sqrt{-5})(a + b\sqrt{-5}) = 2 + \sqrt{-5}$. If $N(a + b\sqrt{-5}) = 1$, then $a + b\sqrt{-5} = \pm 1$, so $\langle a + b\sqrt{-5} \rangle = R$, absurd. Else, $N(a + b\sqrt{-5}) = 9$, so $N(x + y\sqrt{-5}) = N(u + v\sqrt{-5}) = 1$, so $x + y\sqrt{-5}, u + v\sqrt{-5} = \pm 1$, so $a + b\sqrt{-5} = \pm 3$ and $a + b\sqrt{-5} = \pm(2 + \sqrt{-5})$, absurd.

Let R be a PID and $a, b \in R$ be nonzero. Let $\langle d \rangle = \langle a, b \rangle$. Then d is a GCD of a, b , $d = ax + by$ for some $x, y \in R$, and d is unique up to a unit in R .

(1). $a \in \langle d \rangle$, so $d \mid a$. Similarly, $d \mid b$. Suppose $g \mid a, b$. Since $d \in \langle a, b \rangle$, there exist $x, y \in R$ such that $d = ax + by$, so $g \mid d$.

(2). Shown.

(3). Suppose $\langle d \rangle = \langle d' \rangle$. Then there exist $u, v \in R$ such that $d = d'u, d' = dv$, so $d = duv$. Since $d \neq 0$ and R is an integral domain, $uv = 1$, ie $u, v \in R^\times$, AWD.

Let R be an integral domain. Then $R[x]$ is a PID iff R is a field. One direction was shown. Suppose that $R[x]$ is a PID.

Suppose R has a nonzero ideal $I \triangleleft R$. Since $R[x]$ is a PID, let $\langle I, x \rangle = \langle f \rangle$. Since $I \neq 0$, pick nonzero $a \in I$. Since $a \in \langle f \rangle$, $a = fg$ for some $g \in R[x]$. Since a is nonzero constant and R is an integral domain, f, g are also nonzero constant (consider degrees).

Since $x \in \langle f \rangle$, $x = fh$ for some $h \in R[x]$. Let b be the leading coefficient of h . Then $fb = 1$, so f, b are units. Then $\langle f \rangle = R$. Since R is a commutative ring with no nontrivial ideals, R is a field.

Suppose R is an integral domain with a Dedekind-Hasse norm N . Then R is a PID. It remains to check that every ideal $I \triangleleft R$ is principal. If $I = 0$, we're done. Else, $I \neq 0$ so pick N -minimal nonzero $d \in I$. Let $x \in I$. If $x = 0$, we're done. Else, $x \neq 0, d \neq 0$ so by assumption $x \in \langle d \rangle$ (and we're done) or there exist $s, t \in R$ such that $sx - td \neq 0$ and $N(sx - td) < N(d)$. Since $sx - td \in I$, the minimality of $N(d)$ is contradicted. It follows that $I < \langle d \rangle$. Since $d \in I$, we have $\langle d \rangle < I$, so $I = \langle d \rangle$, so I is principal.

3 Quadratic Integers

Let $F = \mathbb{Q}(\sqrt{D})$ be a quadratic field and $\mathcal{O}$ the associated integer ring. Denote by $N : F \rightarrow \mathbb{R}_{\geq 0}$ the associated field norm.

I claim that $\mathcal{O}$ is a Euclidean domain for $D = -1, -2, -3, -7, -11$. I claim that it is enough to show that any field element γ is at most κ away from an element of $\mathcal{O}$ for some fixed $\kappa < 1$, so that $N|_{\mathcal{O}}$ is a Euclidean norm on $\mathcal{O}$.

Suppose $\alpha, \beta \in \mathcal{O}$ with $\beta \neq 0$. Then $\gamma = \alpha\beta^{-1} \in F$. Pick $\delta \in \mathcal{O}$ such that $N(\gamma - \delta) \leq \kappa$. Then $N(\alpha - \delta \cdot \beta) \leq \kappa \cdot N(\beta) < N(\beta)$, since $N(\beta) > 0$. Take $\rho = \alpha - \delta \cdot \beta \in \mathcal{O}$, so $\alpha = \delta \cdot \beta + \rho$ for $N(\rho) < N(\beta)$, as desired.

Consider $D = -1, -2$. Then $\mathcal{O} = \{a + b\sqrt{D} : a, b \in \mathbb{Z}\}$. Suppose $\gamma \in F$. Since $1, \sqrt{D}$ is a basis of $\mathbb{C}$ over $\mathbb{R}$, there exist $x, y \in \mathbb{R}$ such that $\gamma = x + y\sqrt{D}$. Let $x = n + a, y = m + b$ for $n, m \in \mathbb{Z}, a, b \in [0, 1)$, so $\gamma' = \gamma - (n + m\sqrt{D}) = a + b\sqrt{D}$.

Since $n + m\sqrt{D} \in \mathcal{O}$, it suffices to show γ' is at most κ away from an element of $\mathcal{O}$. Consider the four cases $(x, y) \in [0, 1/2) \times [0, 1/2), [0, 1/2) \times [1/2, 1), [1/2, 1) \times [0, 1/2), [1/2, 1) \times [1/2, 1)$.

In the first case, consider the N -distance of γ' to 0: $N(\gamma') = x^2 + |D|y^2 < \frac{1+|D|}{4}$. For the second, consider the distance to $\sqrt{D}$: $N(\gamma' - \sqrt{D}) = x^2 + |D|(1-y)^2 < \frac{1+|D|}{4}$. In the third, $N(\gamma' - 1) = (1-x)^2 + |D|y^2 < \frac{1+|D|}{4}$. In the last, equality is achievable: $N(\gamma' - (1 + \sqrt{D})) = (1-x)^2 + |D|(1-y)^2 \leq \frac{1+|D|}{4}$. It follows that we can take $\kappa = \frac{1+|D|}{4} < 1$ for $D = -1, -2$.

Consider $D = -3, -7, -11$. Then $\mathcal{O} = \{a + b\sqrt{D} : a, b \in \mathbb{Z}\} \cup \{(\frac{1}{2} + \frac{1}{2}\sqrt{D}) + (a + b\sqrt{D}) : a, b \in \mathbb{Z}\}$. Observe that $\mathcal{O}$ is the lattice generated by $u = \frac{1+\sqrt{D}}{2}$ and $v = \frac{1-\sqrt{D}}{2}$, and these elements form a basis of $\mathbb{C}$ over $\mathbb{R}$.

The idea is similar to before. Let $\gamma = xu + yv$, assume $x, y \in [0, 1)$, so γ is contained in a “diamond.” Assume γ is in the top left corner. Shift x, y to s, t so that we have $s, t \in [-1/4, 1/4], t \leq s$ and call this triangular region T . Solve for when $(s - \frac{1}{4})^2 + D(t - \frac{1}{4})^2 \leq (s + \frac{1}{4})^2 + D(t + \frac{1}{4})^2$, or that $t \geq -\frac{1}{4}s$. This divides T into two polygonal regions. By properties of the function $f(x, y) = x^2 + Dy^2$, it suffices to maximize f over the edges of these regions, for which we find $\kappa \leq \frac{(1+|D|)^2}{16|D|}$. For $D = -3, -7, -11$, we find $\kappa < 1$.

Let $I \triangleleft \mathbb{Z}[i]$ be nonzero. Then $\mathbb{Z}[i]/I$ is a finite. Proof: Since $\mathbb{Z}[i]$ is a Euclidean domain, $\mathbb{Z}[i]$ is a PID. Let $I = (\alpha)$ for some $\alpha \neq 0$. Suppose $x \in \mathbb{Z}[i]$. Since $\alpha \neq 0$ by Division algorithm, let $x = q\alpha + r$ with $N(r) < N(\alpha)$. Denote by $S_k = \{x \in \mathbb{Z}[i] : N(x) = k\}$. By properties of $\mathbb{Z}[i]$, S_k is finite. Any element of x can be represented up to α by an element r of $\bigcup_{k \in [N(\alpha)-1]} S_k$, which is finite. Hence, $\mathbb{Z}[i]/I$ is finite.

Let R be a commutative ring and $S, T \subset R$. Let $I = (S), J = (T)$ be ideals. Then $IJ = (ST)$.

Suppose $r = \sum (xs) \cdot (yt) \in IJ$ is decomposable. Then $r = \sum (xy) \cdot (st)$, so $r \in (ST)$. Conversely, suppose $r = \sum x(st) \in (ST)$. Then $r = \sum (xs)t \in IJ$.

Let R be an integral domain such that every prime ideal is principal. Then every ideal of R is principal, ie R is a PID.

Suppose $I \triangleleft R$ is a proper ideal. Denote by $\mathcal{J} = \{J \triangleleft R : I \subset J, J \neq R\}$.

Since $I \in \mathcal{J}$, $\mathcal{J} \neq \emptyset$. Suppose $\mathcal{C} \subset \mathcal{J}$ is a chain. Let $J = \bigcup \mathcal{C}$, so J is an ideal containing I . If J is improper, ie $J = R$, then $1 \in J$, so $1 \in J_\alpha$ for some α , so $J_\alpha = R$, absurd. Hence, $J \in \mathcal{J}$ is a maximum of $\mathcal{C}$. By Zorn's lemma, let $M \in \mathcal{J}$ be maximal, ie there exists a maximal (proper) ideal M containing I .

Since M is maximal, R/M is a field, and therefore an integral domain, so M is a prime ideal. By assumption, M is principal...

A subideal of a nontrivial principal ideal need not be principal. Let $R = \mathbb{Z}[\sqrt{-5}]$. Consider $(4, 2 + 2\sqrt{-5}) \triangleleft (2)$. Suppose FTSOC $(4, 2 + 2\sqrt{-5}) = (a + b\sqrt{-5})$. Because the field norm $N(a + b\sqrt{-5}) = a^2 + 5b^2$ is completely multiplicative and $4, 2 + 2\sqrt{-5} \in (a + b\sqrt{-5})$, we have $a^2 + 5b^2 \mid 16, 24 \rightarrow a^2 + 5b^2 \mid \gcd(16, 24) = 8$. Consider $a^2 + 5b^2 = m$ for $m = 1, 2, 4, 8$. If $m = 1$, then $a + b\sqrt{-5} = \pm 1$, so $4 \mid 1$, absurd. There are no solutions to $m = 2, 8$. When $m = 4$, then $a = \pm 2$, so there exist $\alpha, \beta \in R$ such that $4\alpha + (2 + 2\sqrt{-5})\beta = 2$. Since R is an integral domain, we may cancel, so $2\alpha + (1 + \sqrt{-5})\beta = 1$, which was shown to be impossible.

4 Principal Ideal Domains

Suppose FTSOC there exists a nonprincipal ideal. Denote by $\mathcal{I} \neq \emptyset$ the set of nonprincipal ideals. Let $\mathcal{C} \subset \mathcal{I}$ be a chain, so $I = \bigcup \mathcal{C}$ is an ideal. Suppose that I is principal, so that $I = (a)$ for some $a \in I$, so $a \in I_\alpha$ for some α . Then $(a) < I_\alpha < (a)$ (because I_α is an ideal) so $I_\alpha = (a)$, absurd. Hence, I is nonprincipal and every chain $\mathcal{C}$ has a maximum in $\mathcal{I}$. By Zorn's lemma let M be a maximal nonprincipal ideal.

I claim that M is a prime ideal. Suppose there exist $a, b \notin M$ such that $ab \in M$. Let $I = (M, a, b)$, which has M as a proper ideal. Suppose FTSOC $I = (c)$, so write $a = cx, b = cy$. Suppose $ab = m = cz$. Then $c^2xy = cz \rightarrow cxy = z$ (since $c \neq 0$).

Proof: Let I be an ideal maximal to being nonprincipal. Suppose FTSOC I is not prime, so there exist $a, b \notin I$ such that $ab \in I$. Let $I_a = (I, a)$ and $I_b = (I, b)$. Since $I_a \supsetneq I$, I_a is principal, so $I_a = (\alpha)$ for some α . Denote by $J = \{r \in R : rI_a \subset I\}$, so $I \subsetneq I_b \subset J$, so J is principal, so $J = (\beta)$. Then $I_a J = (\alpha\beta) \subset I$.

Now suppose $x \in I$. Then $x \in I_a$, so there exists $r \in R$ such that $x = r\alpha$. Since $x \in I$, we have $r \in J$, so $x \in I_a J$. It follows $I_a J = I = (\alpha\beta)$ is principal, contradiction.

5 Unique Factorization Domains

Let R be an integral domain with $a, b \neq 0$. Suppose $a \mid b$ and $b \mid a$. Then $b = ua$ for some $u \in R^\times$.

Proof: Write $b = ua, a = vb$ for some $u, v \in R$. Then

$$ab1 = ab = (vb)(ua) = (ab)(uv).$$

Since $a, b \neq 0$, $ab \neq 0$ as well. By cancellation, $uv = 1$, so $u, v \in R^\times$.

Let R be a PID. Suppose $a : \mathbb{Z}^+ \rightarrow R$ is a function such that $a_{n+1} \mid a_n$ and $a_n \neq 0$ for all $n \in \mathbb{Z}^+$. Then there exists an $N \in \mathbb{Z}^+$ such that for $n > N$, $a_n = u_n a_N$ for some $u_n \in R^\times$.

Proof: Consider $I = (a_1, a_2, a_3, \dots)$. Since R is a PID, $I = (\alpha)$ for some $\alpha \in R$. By definition of an ideal, since $\alpha \in I$, $\alpha = \sum_{i \in S} r_i a_i$ for some finite set S . Note that since $I \neq 0$, $\alpha \neq 0$, so $S \neq \emptyset$, so $N = \max S$ is well-defined.

Observe that $\alpha \in (a(S)) \subset (a_1, a_2, \dots, a_N)$, so

$$(\alpha) \subset (a_1, a_2, \dots, a_N) \subset (a_1, a_2, a_3, \dots) \subset (\alpha).$$

It follows that

$$I = (a_1, a_2, a_3, \dots) = (a_1, a_2, \dots, a_N).$$

Since for all $n \leq N$, we have $a_N \mid a_n$, by induction $I = (a_N)$. For $n > N$, we have $a_n \mid a_N$ by construction, and $a_N \mid a_n$ by the previous result. Since $a_n, a_N \neq 0$ it follows $a_n = u_n a_N$ for some $u_n \in R^\times$, as desired.

Every PID is a UFD.

Proof: Let R be a PID. Since R is an integral domain, it suffices to show unique factorization. First, existence.

Suppose $a \neq 0$. If $a \in R^\times$, we're done. Else, I claim that I can find an irreducible divisor of a . Define the sequence (a_n) with terms in R dividing a as follows. Let $a_1 = a$. If a_n is irreducible, we're done. Else, write $a_n = a_{n+1} b_{n+1}$ for $a_{n+1}, b_{n+1} \notin R^\times$, so $a_{n+1} \mid a_n$. Suppose FTSOC that the sequence $a : \mathbb{Z}^+ \rightarrow R$ is infinite. Then by the previous lemma, $a_N = a_{N+1} u$ for some $N \in \mathbb{Z}^+$ and $u \in R^\times$. Then $a_{N+1} u = a_{N+1} b_{N+1}$, so $b_{N+1} = u$, absurd. Hence, we can pick an irreducible factor p of a .

Similarly, we will construct a sequence (c_n) with terms in R dividing a . Let $c_1 = a$. Suppose c_n is defined. If $c_n \in R^\times$, we're done. Else write $c_n = p_n c_{n+1}$ for some irreducible p_n . Suppose FTSOC that the sequence $c : \mathbb{Z}^+ \rightarrow R$ is infinite. Again, we have a contradiction, so we may write $a = p_1 p_2 \cdots p_n$ for

some irreducibles p_i (absorb the last unit into a prime).

Now we show uniqueness. Suppose $a = \prod p_i = \prod q_j$. Since p_1 is irreducible, p_1 is prime. It follows that one of q_1 and $q_2 q_3 \cdots q_m$ is divisible by p_1 . By induction*, at least one of $q_1, q_2, \dots, q_m$ is divisible by p_1 and therefore associate to p_1 . Then divide both sides, so by induction on the number of primes, the result follows.

*I am lazy - and aware that this isn't rigorous.

6 Finitely Generated Modules over PIDs

M be a free module with finite basis over a PID A . Then the rank of the basis is unique.

Suppose $\{e_i\}_{i=1}^n, \{f_j\}_{j=1}^m$ are bases of M . WLOG $m > n$. Then $\begin{bmatrix} f_1 & f_2 & \cdots & f_m \end{bmatrix} = \begin{bmatrix} e_1 & e_2 & \cdots & e_n \end{bmatrix} C$ for some matrix $C \in M_{n \times m}(A)$. Since A is an integral domain, $K = A^{-1}A$ is a field. Embed the columns $c_1, c_2, \dots, c_m$ of C in K^n .

Since K^n is a vector space of dimension n and $m \geq n + 1$, pick not all zero elements k_i such that $\sum k_i c_i = 0$. Since formally, $k_i = (a_i, b_i)$ for some $a_i, b_i \in A$, $b_i \neq 0$, and the sum is finite, we may multiply both sides by $\prod b_i$, so there exists a nonzero combination $\sum r_i c_i = 0$ with $r_i \in A$.

Then

$$\begin{aligned} \begin{bmatrix} f_1 & f_2 & \cdots & f_m \end{bmatrix} \begin{bmatrix} r_1 \\ r_2 \\ \vdots \\ r_m \end{bmatrix} &= \begin{bmatrix} e_1 & e_2 & \cdots & e_n \end{bmatrix} C \begin{bmatrix} r_1 \\ r_2 \\ \vdots \\ r_m \end{bmatrix} \\ &= \begin{bmatrix} e_1 & e_2 & \cdots & e_n \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix} = 0, \end{aligned}$$

contradiction.

If A is a PID and not a field, then A has a prime element. Proof: Let $I \triangleleft A$ be a maximal ideal. Since A is not a field, $I \neq 0$. Since I principal, $I = (\alpha)$ for some $\alpha \neq 0$. Since maximal ideals are prime ideals, α is prime.